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Some problems on systems of equations and integration

Numerical Analysis

Solve the given problems:

1. Solve the following systems $Ax = b$ by Gaussian elimination without pivoting. Check that $A = LU$.

(a) $A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 2 & -2 \\ -2 & 1 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

(b) $A = \begin{bmatrix} 4 & 3 & 2 & 1 \\ 3 & 4 & 3 & 2 \\ 2 & 3 & 4 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$

2. Consider the linear system

$$6x_1 + 2x_2 + 2x_3 = -2$$

$$2x_1 + \left(\frac{2}{3}\right) x_2 + \left(\frac{1}{3}\right) x_3 = 1$$

$$x_1 + 2x_2 - x_3 = 0$$

and verify its solution is $x_1 = 2.6$, $x_2 = -3.8$, $x_3 = -5.0$

3. Solve the following systems of equations

(a) $\begin{bmatrix} 2 & 2 & 1 \\ 4 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 1 & 1 & 2 \\ 4 & 0 & 2 & 1 \\ 3 & 2 & 2 & 0 \\ 1 & 3 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \\ -4 \end{bmatrix}$

using the Gauss elimination method with partial pivoting.

4. Show that the following matrices are non-singular, but they cannot be written as the product of the lower and upper triangular matrices

(a) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 1 & -2 \\ 4 & 2 & 3 \\ -6 & 3 & 7 \end{bmatrix}$

5. Determine the LU factorization of the matrix

$$\begin{bmatrix} 2 & -6 & 10 \\ 1 & 5 & 1 \\ -1 & 15 & -5 \end{bmatrix}$$

so that $(a)\ l_{ii} = 1;$ and $(b)\ u_{ii} = 1$ for $i = 1, 2, 3$.

6. Solve the system of equations

(a) $\begin{bmatrix} 1 & 1 & -1 \\ 2 & 3 & 5 \\ 3 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 1 & -4 & 1 \\ -4 & 3 & 5 & -2 \\ 1 & -1 & 1 & -1 \\ 1 & 3 & -3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 \\ -10 \\ 2 \\ -1 \end{bmatrix}, \ u_{ii} = 1, i = 1(1)4.$

by the LU decomposition method.

7. Solve the system of equations

$\begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ by the Cholesky method.

8. Find the inverses of the matrices

$(a) \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix}$ and $(b) \begin{bmatrix} 1 & -1 & 2 \\ -1 & 4 & 6 \\ 2 & 6 & 29 \end{bmatrix}$

9. Calculate the condition numbers $\operatorname{cond}(A)_p, p = 1, 2, \infty$, for $A = \begin{bmatrix} 100 & 99 \\ 99 & 98 \end{bmatrix}$.

10. Determine the condition number of the matrix using $||.||_1, ||.||_2,$ and $||.||_\infty$

$$A = \begin{bmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 9 & 16 & 25 \end{bmatrix}$$

11. Let $A(\alpha) = \begin{bmatrix} 0.1\alpha & 0.1\alpha \\ 1.0 & 1.5 \end{bmatrix}$. Determine α such that condition number of $A(\alpha)$ is minimized. Use the norm $||.||_\infty$.

12. Find the necessary and sufficient conditions on k, so that the (i) Jacobi method, (ii) Gauss-Seidel method converges for solving the system of equations $Ax = b$

$$A = \begin{bmatrix} 1 & 0 & k \\ 2 & 1 & 3 \\ k & 0 & 1 \end{bmatrix} \text{ and } b \text{ is arbitrary.}$$

13. Given the matrix $A = I + L + U$ where

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 2 & 2 & 1 \end{bmatrix}$$

L and U are strictly lower and upper triangular matrices respectively, decide whether (a) Jacobi and (b) Gauss-Seidel methods converge to the solution $Ax = b$, and b is arbitrary.

14. Show that both the (a) Jacobi and (b) Gauss-Seidel methods diverge for solving the system of equations

$$\begin{bmatrix} 2 & 3 & 1 \\ 3 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 6 \end{bmatrix}$$

15. For the following systems of equations

(a) $4x + y + 2z = 4,\ \ 3x + 5y + z = 7, x + y + 3z = 3$

(b) $10x + 4y - 2z = 12, x - 10y - z = -10, 5x + 2y - 10z = -3$

(i) Show that the Jacobi iteration scheme converges

(ii) Obtain the Jacobi iteration scheme in matrix form

(iii) Starting with $\boldsymbol{x}^{(\mathbf{0})} = \mathbf{0}$, iterate three times.

16. For the following systems of equations

$$(a) \begin{bmatrix} -3 & 1 & 0 \\ 2 & -3 & 1 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ -1 \end{bmatrix} \quad (b) \begin{bmatrix} 5 & 1 & -2 \\ 3 & 4 & -1 \\ 2 & -3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ 10 \end{bmatrix}$$

(i) Set up the Gauss-Seidel iteration scheme in matrix form

(ii) Show that the iteration scheme is convergent and hence find its rate of convergence.

(iii) Starting with $\boldsymbol{x}^{(\mathbf{0})} = \mathbf{0}$, iterate three times.

17. The system of equations $x^2y + y^3 = 10$ and $xy^2 - x^2 = 3$ has a solution near $x = 0.8,\ y = 2.2$. Perform two iterations of the Newton's method to obtain this root.

18. The system of equations

$y\cos(xy)+1=0$, and $\sin(xy)+x-y=0$ has one solution close to $x=1$, $y=2$. Calculate this solution correct to two decimal places.

19. The system of equations $\log(x^2+y)-1+y=0$ and $\sqrt{x}+xy=0$ has one approximate solution $(x_0, y_0)=(2.4,-0.6)$. Improve this solution and estimate the accuracy of the result.

20. Consider the system $x=1+h\,\frac{e^{-x^2}}{1+y^2}$ and $y=0.5+h\tan^{-1}(x^2+y^2)$. Show that if h is chosen sufficiently small, then this system has a unique solution α within some rectangular region. Moreover, show that the simple fixed point iteration $\boldsymbol{X}_{n+1}=\boldsymbol{G}(\boldsymbol{X}_n)$, $\boldsymbol{X}_n=(x_n, y_n)^T$ will converge to this solution.

21. Solve the system $x^2+xy^3=9,\ 3x^2y-y^3=4$ using Newton's methos for nonlinear systems. Use each of the initial guesses $(x_0, y_0)=(1.2, 2.5), (-2, 2.5)$, $(-1.2, -2.5)$, $(2, -2.5)$. Observe which root to which the method converges, the number of iterates required, and the speed of convergence.

22. Verify that the iteration $\boldsymbol{x}^{(k+1)}=\boldsymbol{g}\left(\boldsymbol{x}^{(k)}\right)$, $k=0,1,2, \ldots$, where $g=(g_1, g_2)T$ and g_1 and g_2 are functions of two variables defined by

$g_1(x_1, x_2)=\frac{1}{3}(x_1^2-x_2^2+3)$ and $g_2(x_1, x_2)=\frac{1}{3}(2x_1x_2+1)$, has the fixed point $\boldsymbol{x}=(1,1)^T$.

23. The vector function $\boldsymbol{x}\mapsto\boldsymbol{f}(\boldsymbol{x})$ of two variables is defined by

$f_1(x_1, x_2)=x_1^2+x_2^2-2$, $f_2(x_1, x_2)=x_1-x_2$. Verify that the equations $\boldsymbol{f}(\boldsymbol{x})=\boldsymbol{0}$ has two solutions $x_1=x_2=1$ and $x_1=x_2=-1$. Show that one iteration of Newton's method for the solution of this system gives

$\boldsymbol{x}^{(1)}=\left(x_1^{(1)}, x_2^{(1)}\right)^T$, with $x_1^{(1)}=x_2^{(1)}=\frac{\left(\left(x_1^{(0)}\right)^2+\left(x_2^{(0)}\right)^2+2\right)}{2\left(x_1^{(0)}+x_2^{(0)}\right)}$. Deduce that the iteration converges to $(1,1)^T$ if $x_1^{(0)}+x_2^{(0)}$ is positive and if $x_1^{(0)}+x_2^{(0)}$ is negative the iteration converges to the other solution. Verify that the convergence is quadratic.

24. Consider the simultaneous nonlinear equations

$$f_1(x,y,z) \equiv x^2 + y^2 + z^2 - 1 = 0$$

$$f_2(x,y,z) \equiv 2x^2 + y^2 - 4z = 0$$

$$f_3(x,y,z) \equiv 3x^2 - 4y + z^2 = 0$$

where $\boldsymbol{F} = (f_1, f_2, f_3)^T$ and $\boldsymbol{X} = (x,y,z)^T$. Determine the solution to the equation $\boldsymbol{F}(\boldsymbol{X}) = \boldsymbol{0}$ contained in the first octant $\{(x,y,z) \in \mathbb{R}^3 : x > 0, y > 0, z > 0\}$ in $\mathbb{R}^3$ using Newton method, starting with $(0.5, 0.5, 0.5)^T$.

25. Use the trapezoidal and Simpson's composite rule to find the following integrals and bound on the error

$(a)\ \int_0^1 e^{-x^2}\,dx \quad (b)\ \int_0^1 x^{\frac{5}{2}}\,dx \quad (c)\ \int_{-4}^{4} \frac{1}{1+x^2}\,dx \quad (d)\ \int_0^{2\pi} \frac{1}{(2+\cos(x))}\,dx$

$(e)\ \int_0^{\pi} e^x \cos(4x)\,dx \quad (f)\ \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\cos(x)\ln(\sin x)}{\sin^2 x + 1}\,dx \quad (g)\ \int_0^1 \left(1 + \frac{\sin x}{x}\right)\,dx$

26. Compute the value of the integral $\int_{0.5}^{1.5} \frac{2-2x+\sin(x-1)+x^2}{1+(x-1)^2}\,dx$ with an error less than 10^{-4} (use trapezoidal rule of integration).

27. Find the quadrature formula $\int_0^1 \frac{f(x)}{\sqrt{x(1-x)}}\,dx = \alpha_1 f(0) + \alpha_2 f\left(\frac{1}{2}\right) + \alpha_3 f(1)$ which is exact for polynomials of highest possible degree. Then use the formula on $\int_0^1 \frac{1}{\sqrt{x-x^3}}\,dx$ and compare with the exact value.

28. Write down the errors in the approximation of $\int_0^1 x^4\,dx$ and $\int_0^1 x^5\,dx$ by the trapezium rule and Simpson's rule. Hence find the value of the constant C for which the trapezium rule gives the correct result for the calculation of $\int_0^1 x^5 - Cx^4)\,dx$, and show that the trapezium rule gives a more accurate result than Simpson's rule when $\frac{15}{14} < C < \frac{85}{74}$.

29. Let $p_2(x)$ be the quadratic polynomial interpolating $f(x)$ at $x = 0, h, 2h$. Use This to derive a numerical integration formula I_h for $I = \int_0^{3h} f(x)\,dx$. Use a Taylor series expansion of $f(x)$ to show $I - I_h = \frac{3}{8}h^4 f^{(3)}(0) + O(h^5)$.